

HEALTHCARE DATA CLIENT

Automated Validation Workflow

Healthcare Data Quality Automation and Exception Reporting

Document Control	Detail
Engagement	Jul-Oct 2026
Role	Data Analyst
Primary Stakeholder	Healthcare Analyst
Sources	Excel, CSV and SQL/database
Processing Scale	Approximately 250,000 records per batch
Confirmed Flow	Ingest > Validate > Apply Data Quality Rules > Flag Exceptions > Assign Reason Codes > Generate Exception Output / Report > Review

Workflow status: The high-level automated flow is confirmed. Detailed orchestration, scheduling, infrastructure, exact scripts and the original manual workflow are not retained, so they are not presented as historical facts.

1. Workflow Objective

Provide a repeatable validation process for high-volume healthcare data that identifies records requiring review, preserves failure reasons and produces controlled exception outputs for Healthcare Analyst review.

2. Confirmed Automated Flow

Stage	Workflow Action	Primary Output
1. Ingest	Receive healthcare data from a supported Excel, CSV or SQL/database source.	Governed input population
2. Validate	Prepare the batch for applicable validation checks.	Validation-ready records
3. Apply Data Quality Rules	Execute documented checks for duplicates, missing values, dates, formats, IDs, ranges, codes and cross-field mismatches.	Rule-level validation results
4. Flag Exceptions	Identify records failing one or more applicable rules.	Record-level exception status
5. Assign Reason Codes	Attach a reason code or equivalent failure explanation to each validation failure.	Traceable failure classification
6. Generate Exception Output / Report	Separate records requiring review into an exception file, table or report.	Review-ready exception output
7. Review	Enable targeted Healthcare Analyst review of flagged records.	Reviewed exception population

Core design principle: Records are not silently corrected or discarded simply because they fail validation. Failed records remain traceable and are routed to review.

3. Input Layer

Input Area	Workflow Requirement
Supported sources	Excel, CSV and SQL/database
Data domains	Claims, patients, appointments, providers, billing, transactions, clinical records and insurance data
Batch scale	Approximately 250,000 records per batch
Population control	Establish the governed processed population before calculating exception metrics
Traceability	Retain sufficient source or batch reference to reconcile downstream exceptions

4. Validation Layer

Rule Family	Validation Purpose
Duplicate Detection	Identify records meeting the documented duplicate definition.
Missing-Value Check	Identify null, blank or missing values where completeness is required.
Date Validation	Identify dates that fail the documented validity rule.
Format Validation	Identify values that do not conform to the documented expected format.
Identifier Validation	Identify IDs failing the documented identifier rule.
Range Validation	Identify values outside documented valid bounds.
Code Consistency	Identify values inconsistent with a documented valid code set or mapping.
Cross-Field Validation	Identify records violating a documented relationship between fields.

5. Exception Classification Logic

Condition	Record-Level Result	Rule-Level Result
No applicable rule fails	Non-exception record	No validation failure
One rule fails	One Exception Record	One validation failure
Multiple rules fail	One Exception Record	Multiple validation failures
Exception Rate calculation	Count the affected record once	Do not use rule-failure count as the numerator

Exception Rate: Records failing one or more validation rules / Total records processed x 100. The observed engagement result was 2%.

6. Reason-Code Handling

Control	Workflow Behaviour
Failure identification	Each failed rule produces an identifiable validation result.
Reason assignment	A reason code or equivalent failure explanation is associated with the failure.
Multiple failures	A record may retain multiple reasons when more than one rule fails.
Consistency	The same implemented rule should use consistent reason classification across batches.
Traceability	Reason information remains linked to the affected source record.
Historical boundary	Exact original reason-code labels are not retained.

7. Exception Output

Output Element	Purpose
Source record reference	Connect the exception to the processed source record.
Exception flag	Show that the record requires review.
Validation rule	Identify the failed check.
Reason code / failure explanation	Explain why the record was flagged.
Batch reference	Connect the exception to its processing context.
Relevant review context	Provide only source-supported fields needed for controlled investigation.

8. Reconciliation Controls

Control	Expected Relationship
Input to processed population	Processed records reconcile to the governed batch population.
Processed to exception population	Exception Records are a subset of Records Processed.
Exception to failure output	Every Exception Record has at least one rule-level failure.
Multi-failure handling	Multiple failures do not create multiple record-level exceptions.
Rate reconciliation	Exception Rate recalculates from distinct Exception Records and Records Processed.
Reporting reconciliation	Power BI measures reconcile to the governed validation outputs.

9. Tool Responsibilities

Tool	Supported Workflow Role
Python	Repeatable data profiling, validation and exception-generation logic.
SQL	Data extraction, querying, reconciliation and validation support.
Excel	Source review, spot checks and exception-review support.
Power Query	Reporting-layer preparation and transformation.
Power BI	Data-quality and exception reporting visibility.

Implementation boundary: These tools are confirmed for the engagement. Their exact historical division of responsibility, script design, scheduling method and technical architecture are not retained.

10. Workflow Control Points

Control Point	Control Objective
CP-01 Input population	Prevent unexplained record loss or inflation before validation.
CP-02 Rule applicability	Execute only rules defined for the applicable fields.
CP-03 Parameter governance	Do not infer unretained thresholds, code sets, formats or healthcare relationships.
CP-04 Exception traceability	Keep failed records linked to their source and failure reason.
CP-05 Grain control	Separate record-level exception measures from rule-level failure analysis.
CP-06 Output reconciliation	Reconcile exception output to the processed population.
CP-07 Reporting consistency	Use the same governed results in downstream Power BI reporting.
CP-08 Confidentiality	Exclude PHI and confidential client values from public portfolio outputs.

11. Workflow Testing

Test Area	Expected Check
Ingestion	Supported input enters the workflow and the population is identifiable.
Rule execution	Each implemented rule can generate a pass or fail result.
Exception flagging	Records failing at least one rule are included in the exception population.
Reason codes	Failures carry an identifiable reason or explanation.
Multi-failure records	One record can hold multiple failures without inflating Exception Records.
Exception Rate	The record-level calculation reconciles correctly.
Reporting	Power BI results reconcile to validation and exception outputs.

12. Confirmed Outcome

Outcome Measure	Result
Processing scale	Approximately 250,000 records per batch
Observed Exception Rate	2%
Baseline review turnaround	6 hours per file
Post-solution review turnaround	2.5 hours per file
Time reduction	3.5 hours per file
Derived turnaround reduction	Approximately 58.3%

Outcome boundary: The 58.3% figure describes review turnaround reduction only. It is not presented as labor savings, productivity improvement, clinical-quality improvement or compliance improvement.

13. Workflow Traceability

Workflow Stage	Upstream Definition	Downstream Evidence
Ingest	BRD; Data Dictionary	Data validation testing
Validate	Data Cleaning Methodology	Data Quality Rule Catalogue
Apply rules	Data Quality Rule Catalogue	Exception Analysis
Flag and classify	BRD; User Stories	Exception Analysis
Exception output	Data Dictionary; Rule Catalogue	Testing; Power BI Requirements
Review and report	KPI Catalogue	Findings; Final Report